

The saturation number of 3-term arithmetic progressions in $\{1, \dots, n\}$

Wizard

Let $\text{sat}_3(n)$ be the least size of a maximal 3-AP-free subset of $[n] = \{1, \dots, n\}$. We prove $\text{sat}_3(n) = \Theta(\sqrt{n})$, with explicit constants $\sqrt{2/3} \leq \text{sat}_3(n)/\sqrt{n} < 2$. The upper bound is a placement of Csajbók–Nagy R -complete sets into the interval (no wrap-around). Exact values for $n \leq 40$ are Lean. The spine of the upper bound is formalised in Lean 4 (`lean/`, no `sorry`).

1. Definitions

Let n be a positive integer and write $[n] = \{1, 2, \dots, n\}$. A *3-term arithmetic progression* (3-AP) is a triple of distinct integers x, y, z with $2y = x + z$ in some order. A set $A \subseteq [n]$ is *3-AP-free* if it contains no 3-AP.

The set A is *maximal* 3-AP-free in $[n]$ if it is 3-AP-free and, for every $t \in [n] \setminus A$, the set $A \cup \{t\}$ contains a 3-AP. Equivalently, every missing point is the third vertex of a 3-AP whose other two vertices lie in A .

Write $\text{sat}_3(n)$ for the least cardinality of a maximal 3-AP-free subset of $[n]$. This is the independent domination number of the 3-uniform hypergraph of 3-APs on vertex set $[n]$.

The function is distinct from $r_3(n)$, the maximum size of a 3-AP-free subset of $[n]$ (OEIS A003002). It is also distinct from the minimum size of a complete 3-AP-free set in a cyclic group or vector space, studied by Csajbók and Nagy [1, 2]. Those papers treat $\mathbb{Z}_m$ and $\mathbb{F}_p^n$, not the interval $[n]$. A sequence search on OEIS for

1, 2, 2, 2, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 5, 6, 6, 6, 6, 6

returns no match (checked 2026-09-07). Savchev–Chen [3] treat uniqueness of large 3-AP-free subsets of $[n]$, not the minimal maximal ones. Chen [5] and Fang [6], following Kiss–Sándor–Yang [7], study infinite AP_3 -covering sequences in $\mathbb{N}$: every large enough integer is the third term of a 3-AP with two earlier terms in the sequence. Those sequences need not be 3-AP-free, so the problem is different.

On $\mathbb{Z}$, the relation $c = 2a - b$ with $a \neq b$ forces $c \neq a$ and $c \neq b$. Thus a finite set is 3-AP-free if and only if it is $(2, -1)$ -avoiding in the sense of [1, 2]. Completeness in $[n]$ still differs from completeness in $\mathbb{Z}_n$: there is no wrap-around.

The results below that are not labelled computational are proved in ordinary mathematics. The spine of the upper bound is formalised in Lean 4 (`lean/`, no `sorry`); see §11. Exact values of $\text{sat}_3(n)$ for $n \leq 40$ are Lean (`sat3_eq_*`).

Theorem 1.1. For every integer $n \geq 1$,

$$\text{sat}_3(n) < 2\sqrt{n}.$$

In particular $\text{sat}_3(n) = \Theta(\sqrt{n})$.

The lower bound of matching shape is Proposition 2.2. The proof of the upper bound is Theorem 9.2.

2. Pair bound

Each unordered pair in a 3-AP-free set determines at most three third vertices: $2b - a$, $2a - b$, and (when $a + b$ is even) $(a + b)/2$. A set of size k therefore blocks at most $k + 3\binom{k}{2}$ points of $[n]$, counting the set itself.

Lemma 2.1. For $k \geq 1$,

$$k + 3\binom{k}{2} = \frac{k(3k - 1)}{2}.$$

In particular $2(k + 3\binom{k}{2}) = k(3k - 1)$.

Proof. Direct expansion: $k + 3k(k - 1)/2 = (2k + 3k^2 - 3k)/2 = (3k^2 - k)/2$. The identity $2 \cdot \text{pairBound}(k) = k(3k - 1)$ is `pairBound_double` in `lean/Sat3/PairBound.lean`.

Proposition 2.2. If $A \subseteq [n]$ is maximal 3-AP-free and $|A| = k$, then

$$n \leq \frac{k(3k - 1)}{2}.$$

Hence $\text{sat}_3(n) \geq \min\{k : k(3k - 1)/2 \geq n\}$. In particular $\text{sat}_3(n) = \Omega(\sqrt{n})$, and more precisely

$$\text{sat}_3(n) \geq \sqrt{\frac{2n}{3}}.$$

Proof. Every point of $[n]$ is either in A or a third vertex of a pair from A (`coverList_covers`). The covering list has length `pairBound(k)` (`coverList_length`), so $n \leq \text{pairBound}(k)$ (`pairBound_of_maximal`, and `HasMaximalOfSize_pairBound` for the existence encoding). The identity $k(3k - 1)/2 = \text{pairBound}(k)$ is `pairBound_half`. If $n \leq \text{pairBound}(k)$ then $2n \leq 3k^2$ (`pairBound_quadratic`), hence $k \geq \sqrt{2n/3}$.

The bound is not always tight: $\text{sat}_3(5) = 3$ while the pair bound only forces $k \geq 2$, and $\text{sat}_3(18) = 6$ while the pair bound only forces $k \geq 4$.

3. The 4-point block

For $a \in \mathbb{Z}$ write

$$S(a) = \{a, a + 1, a + 4, a + 5\}.$$

Lemma 3.1. $S(a)$ is 3-AP-free.

Proof. Translate to $S(0) = \{0, 1, 4, 5\}$ and inspect the four-point set. Formalised as `S_AP3Free`.

Lemma 3.2. The pair-completions of $S(a)$ fill the interval $[a - 5, a + 10]$. Explicitly, the third vertices of pairs from $S(0)$ are

$$\{-5, -4, -3, -2, -1, 2, 3, 6, 7, 8, 9, 10\},$$

so $S(0)$ together with its completions is $\{-5, -4, \dots, 10\}$.

Proof. The six pairs and their third vertices:

- $\{0, 1\}$: $-1, 2$
- $\{0, 4\}$: $-4, 2, 8$
- $\{0, 5\}$: $-5, 10$
- $\{1, 4\}$: $-2, 7$
- $\{1, 5\}$: $-3, 3, 9$
- $\{4, 5\}$: $3, 6, 9$

Union with $\{0, 1, 4, 5\}$ is $\{-5, \dots, 10\}$. Formalised as `S0_covers` and `S_covers_interval`.

Corollary 3.3. $S(6) = \{6, 7, 10, 11\}$ is a maximal 3-AP-free subset of $[16]$. Hence $\text{sat}_3(16) \leq 4$. Combined with Proposition 2.2 (which forces $k \geq 4$, since $\text{pairBound}(3) = 12 < 16$) one has $\text{sat}_3(16) = 4$.

Formalised as `S6_maximal_16`, `sat3_exact_16`, and `sat3_eq_sixteen`. The same pin gives `sat3_eq_one..sat3_eq_four` and `sat3_eq_thirteen..sat3_eq_fifteen`.

Lemma 3.4. Let $d \in \{11, 12, 13, 14, 15, 16\}$. Then $S(0) \cup S(d)$ is 3-AP-free, and its pair-completions fill the interval $[-d-5, d+10]$ with no holes.

Proof. A 3-AP with two points in $S(0)$ and one in $S(d)$ would force $d \in 2S(0) - S(0) - S(0) = [-10, 10]$, contradicting $d \geq 11$. The same holds with the two blocks swapped. Completions of mixed pairs fill the gap between the two 16-point intervals $[-5, 10]$ and $[d-5, d+10]$ precisely when $11 \leq d \leq 16$. Boolean certificates `twoBlock_covers_11` through `twoBlock_covers_16` confirm there are no holes. The $d = 16$ case after the translation $a \mapsto a + 6$ is also `twoBlocks_maximal_32`.

Corollary 3.5. $S(6) \cup S(22) = \{6, 7, 10, 11, 22, 23, 26, 27\}$ is a maximal 3-AP-free subset of $[32]$. Hence $\text{sat}_3(32) \leq 8$. Formalised as `twoBlocks_maximal_32`. The exact value is 7 (Theorem 4.1); the two-block set is not a minimiser.

Three copies of $S(\cdot)$ at distance 16 need not be 3-AP-free: $S(0) \cup S(16) \cup S(32)$ contains the 3-AP $0, 16, 32$.

4. Exact values

Theorem 4.1. The values of $\text{sat}_3(n)$ for $1 \leq n \leq 40$ are

$$\begin{aligned} n : & \quad 1, 2, 3, 4, 5, 6-16, 17, 18-26, 27-32, 33-40 \\ \text{sat}_3(n) : & \quad 1, 2, 2, 2, 3, 4, 5, 6, 7, 8. \end{aligned}$$

Explicitly:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$\text{sat}_3(n)$	1	2	2	2	3	4	4	4	4	4	4	4	4	4	4	4	5	6	6	6

n	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
$\text{sat}_3(n)$	6	6	6	6	6	6	7	7	7	7	7	7	8	8	8	8	8	8	8	8

Certificate. For $n = 1$ through 40 the value is Lean (`sat3_eq_*`). Pair-bound pin, two- and three-point enumerations, four-point enumeration at $n = 17$, increasing 4- and 5-tuples at $n = 18-22$, increasing 5-tuples at $n = 23-26$ (`anyMaximalInc`; `pairBound(4)=22` kills size 4 once $n \geq 23$), 3-AP-free 5- and 6-tuples at $n = 27-32$ (`anyMaximalFree`), and a recursive 3-AP-free search at $n = 33-40$ (`anyMaximalSearch`; no maximal 5-, 6-, or 7-set, with `pairBound(5)=35` killing size 5 once $n \geq 36$).

An explicit witness for each n is in `data/sat3_values.json`. `scripts/verify_table.py` rechecks that the witness is maximal of the claimed size and that no 3-AP-free set of size `pairBound` through $\text{sat}_3(n) - 1$ is maximal. For $n \leq 26$ the number of distinct minimisers was also counted (see §5).

The search is deterministic and small: $n = 26$ takes about one second, $n = 40$ about one minute, in CPython, deciding membership of an increasing k -tuple by forbidding midpoints already present. Computationally (not Lean), the same backtrack gives $\text{sat}_3(n) = 8$ for $n = 41$ – 46 , with witnesses in `data/sat3_values_ext.json`. The values $n = 47, 48$ were not finished.

One witness per n :

$n = 1: \{1\},$ $n = 2: \{1, 2\},$
 $n = 9: \{1, 2, 4, 5\},$ $n = 16: \{6, 7, 10, 11\},$
 $n = 18: \{1, 2, 4, 8, 9, 11\},$ $n = 26: \{8, 10, 11, 15, 17, 18\},$
 $n = 27: \{1, 2, 5, 11, 12, 14, 18\},$ $n = 32: \{2, 3, 14, 15, 17, 22, 23\}$ (size 7; $S(6) \cup S(22)$ is a non-minimal maximal)
 $n = 33: \{1, 2, 4, 5, 13, 14, 16, 17\},$ $n = 40: \{1, 2, 5, 6, 16, 17, 20, 21\}.$

Several of these are (translates of) one or two copies of $S(a)$, sometimes with two extra points at the left end.

Remark 4.2. The unique minimiser in [16] is $S(6)$. For $n = 17$ there are exactly two minimisers, $\{1, 7, 8, 11, 12\}$ and $\{6, 7, 10, 11, 17\}$: the unique 4-point set of [16], with either endpoint of [17] adjoined. Neither adjunction is forced by Lemma 3.2, which only covers up to 16.

5. Counts of minimisers

Let $m(n)$ be the number of maximal 3-AP-free subsets of $[n]$ of size $\text{sat}_3(n)$. Exhaustive enumeration gives, for $n \leq 26$:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$m(n)$	1	1	3	1	2	4	10	25	34	24	21	12	7	4	2	1
n	17	18	19	20	21	22	23	24	25	26						
$m(n)$	2	154	146	104	78	46	26	13	6	2.						

This sequence is also absent from OEIS (checked 2026-09-07). The spike at $n = 18$, where sat_3 jumps from 5 to 6, is the first n at which no 5-point maximal set exists; many 6-point configurations then become available.

6. Endpoint covering

Call t an *endpoint completion* of A if $t = 2y - x$ for distinct $x, y \in A$ with $t \notin \{x, y\}$. Write A *endpoint-covers* $[lo, hi]$ if every integer of that interval lies in A or is an endpoint completion of A . Endpoint covering of $[n]$ implies maximality in $[n]$ (midpoints are not required).

Lemma 6.1. Let $d \in \mathbb{Z}$. If A endpoint-covers $[lo, hi]$, then $A + d$ endpoint-covers $[lo + d, hi + d]$.

Proof. If $t \in A$ then $t + d \in A + d$. If $t = 2y - x$ then $t + d = 2(y + d) - (x + d)$. Formalised as `endpoint_covers_translate`.

Lemma 6.2. Let $A \subseteq \mathbb{Z}$ be 3-AP-free. Then $A + d$ is 3-AP-free. Formalised as `AP3Free_shift`.

7. R -complete sets

The following definition is due to Csajbók–Nagy [2, Definition 3.1], who used it to produce complete $(2, -1)$ -avoiding subsets of $\mathbb{Z}_m$. We use it on the interval $[n]$, with a placement that does not wrap around.

Definition 7.1. Let R be a nonnegative integer. A set $A \subseteq [0, R]$ is R -complete if it is 3-AP-free and its endpoint completions fill $[-R, 2R]$.

Write $I(A) = [2 \min A - \max A, 2 \max A - \min A]$ for the far-endpoint interval of a finite set A . If $A \subseteq [0, R]$ and $0, R \in A$, then $I(A) = [-R, 2R]$.

Lemma 7.2 (placement, left). Let A be R -complete and let n satisfy $R + 1 \leq n \leq 2R + 1$. Then $A + 1$ is a maximal 3-AP-free subset of $[n]$.

Proof. The points of $A + 1$ lie in $[1, R + 1] \subseteq [n]$. Freeness is Lemma 6.2. Lemma 6.1 translates the covering of $[-R, 2R]$ to $[1 - R, 2R + 1]$. The interval $[1, n]$ is contained in $[1 - R, 2R + 1]$ because $n \leq 2R + 1$. Formalised as `rcomplete_place_left`.

Lemma 7.3 (placement, right). Let A be R -complete and let n satisfy $2R + 1 \leq n \leq 3R + 1$. Then $A + (n - 2R)$ is a maximal 3-AP-free subset of $[n]$.

Proof. Write $s = n - 2R$, so $s \geq 1$. The points of $A + s$ lie in $[s, R + s] = [n - 2R, n - R] \subseteq [n]$. Freeness is Lemma 6.2. Lemma 6.1 translates the covering of $[-R, 2R]$ to $[n - 3R, n]$. The hypothesis $n \leq 3R + 1$ gives $n - 3R \leq 1$, so $[1, n] \subseteq [n - 3R, n]$. Formalised as `rcomplete_place_right`.

Corollary 7.4. If an R -complete set of size k exists and $R + 1 \leq n \leq 3R + 1$, then $\text{sat}_3(n) \leq k$.

Proof. If $n \leq 2R + 1$ use Lemma 7.2; if $n \geq 2R + 1$ use Lemma 7.3. At $n = 2R + 1$ the two shifts coincide. Formalised as `rcomplete_place`.

The modular analogue in [2, Lemma 3.2] requires the stricter window $2R < m \leq 3R + 1$, because wrap-around is used to cover the left end. On the interval the left end is covered by a unit shift, so the window is longer.

8. Admissible digits

The next two statements are the interval form of [2, Lemma 3.3 and Corollary 3.5]. The proofs are written out so that they can be checked without the group-theoretic paper. The inductive covering step is `rcomplete_admissible_step`. The greedy digit sequence for every $R \in I_k$ is `greedyDigits_admissible`.

Lemma 8.1. Let $A \subseteq [0, S]$ be S -complete, with $S \geq 0$, and let d be an integer satisfying $2S + 1 \leq d \leq 3S + 1$. Then $A \cup (A + d)$ is $(S + d)$ -complete.

Proof. Write $U = A \cup (A + d)$. The points of U lie in $[0, S + d]$.

Freeness. The set A lies in $[0, S]$ and $A + d$ lies in $[d, S + d]$ with $d \geq 2S + 1$, hence $2 \max A = 2S < d \leq \min(A + d)$. If a 3-AP has two points in A and one in $A + d$, the point of $A + d$ is either $2y - x$ with $x, y \in A$, hence at most $2S$, or the middle of two points of A , hence at most S ; both contradict membership in $A + d$. If it has two points in $A + d$ and one in A , the point of A is either $2u - v$ with $u, v \in A + d$, hence at least $2d - (S + d) = d - S \geq S + 1$, or the middle of two points of $A + d$, hence at least d ; both lie to the right of A . Formalised as `separated_union_AP3Free`.

Covering. Write $I = [-S, 2S]$. The endpoint completions of U include the four families

$$I - d, \quad I, \quad I + d, \quad I + 2d,$$

namely $2A - (A + d)$, the given covering of I , its translate by d , and $2(A + d) - A$. Distinctness of mixed triples uses $d > 2S$ as in the freeness paragraph (formalised as `mixed_left_I` and `mixed_right_I`).

These four intervals abut or overlap because $d \leq 3S + 1$: the right end of I is $2S$ and the left end of $I + d$ is $d - S$, and $d - S \leq 2S + 1$ is exactly $d \leq 3S + 1$. The same comparison joins $I - d$ to I and $I + d$ to $I + 2d$. Their union is $[-(S + d), 2(S + d)]$. Formalised as `admissible_shift_covers`.

Lemma 8.2. Let $d_1 = 1$ and $S_i = d_1 + \dots + d_i$. Suppose $2S_i + 1 \leq d_{i+1} \leq 3S_i + 1$ for each $i = 1, \dots, k - 1$. Let

$$P = \left\{ \sum_{i=1}^k \varepsilon_i d_i : \varepsilon_i \in \{0, 1\} \right\}.$$

Then P is an S_k -complete set of size 2^k . Covering and freeness of the nested union $A \cup (A + d_2) \cup \dots$ are Lean (`rcomplete_from_admissible`, iterating `rcomplete_admissible_step` from $\{0, 1\}$). Distinctness of the nested doubling is Lean (`nodup_growFrom`): $P \subseteq [0, S]$ and $d \geq 2S + 1$ force P and $P + d$ to be disjoint, so the grown list is duplicate-free. The greedy nested union is exactly that list (`greedy_set_iff`) of length 2^k (`greedy_growFrom_length`).

Proof. Distinctness. Nested doubling: if $P \subseteq [0, S]$ is duplicate-free and $d \geq 2S + 1$, then P and $P + d$ are disjoint, so $|P \cup (P + d)| = 2|P|$. Formalised as `nodup_growFrom`. (Equivalently, a nonempty signed sum $\sum \delta_i d_i = 0$ with $\delta_i \in \{-1, 0, 1\}$ is impossible: the largest nonzero coefficient has size $d_j > S_{j-1}$.)

Freeness and covering. Iterate Lemma 8.1 from $\{0, 1\}$. Formalised as `rcomplete_from_admissible`. The greedy nested union is exactly the grown list (`greedy_set_iff`) of length 2^k (`greedy_growFrom_length`).

Write

$$I_k = \left[\frac{3^k - 1}{2}, \frac{4^k - 1}{3} \right] \cap \mathbb{Z},$$

and $\ell_k = (3^k - 1)/2$, $u_k = (4^k - 1)/3$, so $I_k = [\ell_k, u_k]$.

Corollary 8.3. For every $k \geq 1$ and every integer $R \in I_k$ there exists an R -complete set of size 2^k .

Proof. Construct digits by downward induction. For $k = 1$, $I_1 = \{1\}$ and $d_1 = 1$. Given $R \in I_{k+1}$, set

$$S = \min(u_k, \lfloor (R - 1)/3 \rfloor), \quad d = R - S.$$

Then $S \in I_k$ and $2S + 1 \leq d \leq 3S + 1$. Indeed $S \leq u_k$ by construction, while $R \geq \ell_{k+1} = 3\ell_k + 1$ forces $\lfloor (R - 1)/3 \rfloor \geq \ell_k$, hence $S \geq \ell_k$. The lower digit bound is $3S \leq R - 1$. For the upper bound: if $S = u_k$ then $R \leq u_{k+1} = 4u_k + 1 = 4S + 1$; if $S = \lfloor (R - 1)/3 \rfloor$ then $R \leq 3S + 3$, and $3S + 3 \leq 4S + 1$ once $S \geq 2$. The remaining case $k = 1$ has $I_2 = \{4, 5\}$ and $S = 1 = u_1$, so it falls in the first alternative. Formalised as `greedyPred_spec` (`k=1` by computation on $I_2 = \{4, 5\}$; `k \geq 2` from $\ell_k \geq 4$). The inductive hypothesis supplies an admissible sequence for S ; adjoining d yields one for R . Formalised as `greedyDigits_admissible`. Iterating the one-step covering along that list yields an R -complete set (`greedy_rcomplete`). The nested union is a duplicate-free list of length 2^k (`greedy_growFrom_nodup`, `greedy_growFrom_length`, `greedy_set_iff`). Implemented in `scripts/rcomplete.py`.

(The same recurrence also shows that the attainable sums fill I_k : each $S \in I_k$ contributes the block $[3S + 1, 4S + 1]$, consecutive blocks meet once $S \geq 2$ because $3S + 4 \leq 4S + 2$

(consecutive_images_overlap), and after $k = 1$ one has $S \geq 4$. This overlapping-union picture is not used below.)

The extreme choice $d_i = 4^{i-1}$ recovers the base-4 subset sums of Csajbók–Nagy [1, 2]: then $S_k = (4^k - 1)/3$ and $d_{k+1} = 3S_k + 1$.

9. Every n

Lemma 9.1. Let $n \geq 2$ and let $k \geq 1$ be the integer with $4^{k-1} < n \leq 4^k$. Then $R = \max(\ell_k, \lceil (n-1)/3 \rceil)$ lies in I_k and satisfies $R+1 \leq n \leq 3R+1$.

Proof. The comparison $\ell_k \leq u_k$ is `Ilo_le_Ihi`. The identity $4^k = 3u_k + 1$ is `four_pow_Ihi`, so $n \leq 4^k$ gives $\lceil (n-1)/3 \rceil \leq u_k$. Thus $R \leq u_k$. The identity $3^k = 2\ell_k + 1$ is `three_pow_Ilo`. The inequality $3^k \leq 2 \cdot 4^{k-1} + 1$ is `lower_window`, hence $\ell_k + 1 \leq 4^{k-1} + 1 \leq n$. Also $\lceil (n-1)/3 \rceil + 1 \leq n$ for $n \geq 2$. Thus $R+1 \leq n$. Finally $3\lceil (n-1)/3 \rceil \geq n-1$, so $n \leq 3R+1$. Formalised as `chosenR_window`.

Theorem 9.2. For every integer $n \geq 2$,

$$\text{sat}_3(n) \leq 2^k < 2\sqrt{n},$$

where k is defined by $4^{k-1} < n \leq 4^k$. For $n = 1$ one has $\text{sat}_3(1) = 1 < 2$.

Proof. Lemma 9.1 supplies $R \in I_k$ with $R+1 \leq n \leq 3R+1$. Corollary 8.3 supplies an R -complete set of size 2^k . Corollary 7.4 places it into $[n]$, so $\text{sat}_3(n) \leq 2^k$. Since $n > 4^{k-1}$ one has $(2^k)^2 = 4^k = 4 \cdot 4^{k-1} < 4n$, hence $2^k < 2\sqrt{n}$. Formalised as `sat3_theta` for every $n \geq 1$: `HasMaximalOfSize n (sat3Upper n)` with `sat3Upper 1 = 1` and `sat3Upper n = 2^k` for $n \geq 2$, together with $(2^k)^2 < 4n$ when $n \geq 2$, and $2n \leq 3k^2$ for every maximal set of size k . The construction is `sat3_construction`. The numerical function is the Lean `sat3 n (sat3_spec, sat3_le_upper, sat3_sq_lt)`.

Theorem 1.1 is Theorem 9.2 together with Proposition 2.2. Formalised as `sat3_theta_num`: $\text{sat}_3(n) \leq \text{sat3Upper}(n)$, $2n \leq 3\text{sat}_3(n)^2$, and $\text{sat}_3(n)^2 < 4n$ when $n \geq 2$. The one-step inequality is `sat3_succ_le`. Exact values through $n = 40$ are `sat3_eq_one..sat3_eq_forty`.

Lemma 9.4. $\text{sat}_3(n+1) \leq \text{sat}_3(n) + 1$.

Proof. Let A be maximal 3-AP-free in $[n]$. If $n+1$ is already an endpoint completion of A , then A remains maximal in $[n+1]$. Otherwise $A \cup \{n+1\}$ is 3-AP-free (a 3-AP through $n+1$ would be an endpoint completion) and covers $[n+1]$. Formalised as `maximal_extend_right`, `HasMaximalOfSize_succ`, and `sat3_succ_le`.

Corollary 9.5. For $k \geq 1$ and $t \geq 0$,

$$\text{sat}_3(4^k + t) \leq 2^k + t.$$

In particular $\text{sat}_3(5) \leq 3$ and $\text{sat}_3(17) \leq 5$. Formalised as `sat3_le_interp`, `sat3_le_five`, `sat3_le_seventeen_num`. The pair bound at $n = 5$ only forces $k \geq 2$ (`pairBound 2=5`). Exactness $\text{sat}_3(5) = 3$ is `sat3_eq_five`.

Walking t steps from 4^k is linear. It improves the construction 2^{k+1} only for $t < 2^k$, a $\Theta(\sqrt{n})$ neighbourhood of each 4^k , and does not improve the global constant 2 in Theorem 9.2.

Remark 9.6. The implicit constants are $\sqrt{2/3} \leq \text{sat}_3(n)/\sqrt{n} < 2$. Along $n = 4^k$ the construction uses $2^k = \sqrt{n}$ points, so the upper constant 2 is an artefact of the worst n just after a power of

4. For $n \leq 40$ the construction size equals $\text{sat}_3(n)$ at every $n = 4^k$ in the table ($n = 1, 4, 16$), and also throughout $n = 6..16$ and $n = 33..40$. Just after each power of 4, Corollary 9.5 is tighter than Theorem 9.2 ($n = 5, 17$). No attempt is made here to optimise the constant.

10. A weaker Cantor construction

The following statements were the previous upper bound. They are correct and are recorded for comparison. They are not used in Theorem 9.2.

Let $T = \{0, 1, 4, 5\}$ and let B be the nonnegative integers whose ternary digits lie in $\{0, 1\}$. Let $\mathcal{A} = \bigcup_{b \in B} S(6 + 11b)$. Then $\mathcal{A}$ is 3-AP-free: a 3-AP rearranges to $2t - t' - t'' = 11(b' + b'' - 2b)$ with left side in $[-10, 10]$, so both sides vanish, and B and T are degenerate. Write $B_k = B \cap [0, 3^k)$ and $\mathcal{A}_k = \bigcup_{b \in B_k} S(6 + 11b)$, $L_k = 11 \cdot 3^k + 5$.

If $A \subseteq [1, d + 5]$ endpoint-covers $[1, d + 5]$ and $d > 0$, then $A \cup (A + d)$ endpoint-covers $[1, 3d + 5]$ (formalised as `mixed_fills_right`). The Cantor recurrence $\mathcal{A}_{k+1} = \mathcal{A}_k \cup (\mathcal{A}_k + 11 \cdot 3^k)$ therefore yields that $\mathcal{A}_k$ is maximal 3-AP-free in $[L_k]$, hence $\text{sat}_3(L_k) \leq 4 \cdot 2^k = O(L_k^{\log_3 2})$. A one-point repair at the two residues $n = c(n) + 3, c(n) + 4$ of the last centre extends the bound to every n , still of order $n^{\log_3 2}$.

Base-4 centres are the right-hand extreme of Lemma 8.2 and already give $O(\sqrt{n})$ on each interval $[\max \mathcal{A}_k^{(4)}, 2 \max - \min]$, a positive-density set of n . The complementary gaps are of length $\Theta(4^k)$. Those gaps are filled by the variable digits of Lemma 8.2, not by a one-point repair.

11. What is formalised

Lean 4 project `lean/`, no `mathlib`, no `sorry`.

Proved:

- `S_AP3Free`, `S0_covers`, `S_covers_interval`, `S6_maximal_16`, `twoBlocks_maximal_32`, `twoBlock_covers_11..16`
- `pairBound_double`, `pairBound_half`, `pairBound_quadratic`, `pairBound_of_maximal`, `HasMaximalOfSize_pairBound`
- `T_degenerate`
- `endpoint_covers_translate`, `mixed_fills_right`
- `IsRComplete`, `rcomplete_01`
- `separated_union_AP3Free`, `mixed_left_I`, `mixed_right_I`, `admissible_shift_covers`, `rcomplete_admissible_step`
- `rcomplete_place_left`, `rcomplete_place_right`
- `Ilo`, `Ihi`, `Ilo_le_Ihi`, `four_pow_Ihi`, `three_pow_Ilo`, `lower_window`
- `two_pow_sq_lt`, `chosenR_window`
- `greedyPred_spec`, `admissible_sum_in_I`, `consecutive_images_overlap`
- `greedyDigits`, `greedyDigits_length`, `greedyDigits_sum`, `greedyDigits_last_admissible`, `greedyDigits_admissible`
- `rcomplete_from_admissible`, `greedy_rcomplete`, `greedy_subsetSums_length`, `buildFrom_spec`, `greedy_support`
- `nodup_growFrom`, `greedy_growFrom_nodup`, `greedy_growFrom_length`, `greedy_set_iff`
- `rcomplete_place`, `placedGreedy_maximal`, `placedGreedy_iff`, `windowK_spec`, `sat3_construction`, `sat3_upper`, `HasMaximalOfSize`, `HasMaximalOfSize_one`, `sat3Upper`, `sat3_0_sqrt`, `sat3_theta`

- `pairBound_mono`, `HasMaximalOfSize_ge`, `sat3_exact_when_pin`, `sat3_exact_1..4`, `sat3_exact_13..16`
- `maximal_extend_right`, `HasMaximalOfSize_succ`, `Sat3Le`, `Sat3Le_succ`, `sat3_interp`, `sat3_le_five`, `sat3_le_seventeen`
- `not_maximal_pair_five`, `not_HasMaximalOfSize_5_2`, `sat3_exact_5`
- `exists_least_maximal`, `sat3_min`, `sat3_min_0_sqrt`, `sat3`, `sat3_spec`, `sat3_le_upper`, `sat3_sq_lt`, `sat3_succ_le`, `sat3_theta_num`
- `sat3_eq_one..sat3_eq_forty`, `sat3_le_interp`, `sat3_le_seventeen_num`
- `anyMaximal3_6..12`, `anyMaximal4_17`, `anyMaximalInc` (increasing tuples), `anyMaximalFree` (3-AP-free tuples), `anyMaximalSearch` (DFS), `sat3_eq_eighteen..forty`
- boolean certificates `cert_n*` and `maximal_of_cert_n*` for the table witnesses of Theorem 4.1. The checker is equivalent to maximality: `maximal_iff` (`maximal_sound` and `maximal_complete`)

Not formalised (paper proofs, marked above):

- a computable closed form for $\text{sat}_3(n)$. Lean has a noncomputable `sat3 n`. Exact values through $n = 40$ are Lean (`native_decide` on the enumerators above).
- the Cantor arguments of §10, except `mixed_fills_right`

References

- [1] B. Csajbók and Z. L. Nagy, Complete 3-term arithmetic progression free sets of small size in vector spaces and other abelian groups, arXiv:2401.06283.
- [2] B. Csajbók and Z. L. Nagy, Small complete 3-term progression free sets in cyclic groups and vector spaces, arXiv:2606.30186.
- [3] S. Savchev and F. Chen, A note on maximal progression-free sets, Discrete Math. 306 (2006), 2131–2133.
- [4] OEIS Foundation, Sequence A003002, $r_3(n)$; Sequence A003278, integers with ternary digits in $\{0, 1\}$.
- [5] Y. G. Chen, On AP_3 -covering sequences, C. R. Math. 356 (2018), 121–124.
- [6] J. H. Fang, A note on AP_3 -covering sequences, Period. Math. Hungar. 83 (2021), 67–70.
- [7] S. Z. Kiss, C. Sándor, and Q.-H. Yang, On generalized Stanley sequences, arXiv:1710.01939. They introduce AP_3 -covering sequences (saturating, not necessarily 3-AP-free).